ORDINARY ABELIAN VARIETIES OVER A FINITE FIELD

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We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbf{F}_q$. The result was inspired by Ihara [2], Chapter V; see also [3].

1. Conventions

Let p be a prime number, let $\mathbf{F}_p = \mathbf{Z}/(p)$, let $\overline{\mathbf{F}}_p$ be an algebraic closure of $\mathbf{F}_p$, and, for every power q of p, let $\mathbf{F}_q$ be the subfield of $\overline{\mathbf{F}}_p$ with q elements. For every algebraic extension k of $\overline{\mathbf{F}}_p$, let $W_0(k)$ be the henselian discrete valuation ring essentially of finite type over $\mathbf{Z}$, absolutely unramified, with residue field k. Let $W(k)$ be the ring of Witt vectors over k, that is, the completion of $W_0(k)$; put $W = W(\overline{\mathbf{F}}_p)$, and choose an embedding

$$\varphi : W \hookrightarrow \mathbf{C}.$$

Let $\mathbf{Z}(1)$ denote the subgroup $2\pi i\,\mathbf{Z}$ of $\mathbf{C}$. The exponential map defines an isomorphism

$$\mathbf{Z}(1) \otimes \mathbf{Z}_l \simeq \mathbf{Z}_l(1)(\mathbf{C}) = \varprojlim \mu_{l^n}(\mathbf{C}).$$

For an abelian variety A, write A^* for its dual. For every field k, write $\bar{k}$ for an algebraic closure.

2. Ordinary abelian varieties

Let A be an abelian variety of dimension g over a field k of characteristic p. Recall that A is called *ordinary* if the following equivalent conditions hold:

(I) A has p^g points over $\bar{k}$ of order dividing p.

(II) The Hasse–Witt matrix

$$F^* : H^1(A^{(p)}, \mathscr{O}_{A^{(p)}}) \to H^1(A, \mathscr{O}_A)$$

is invertible.

(III) The identity component of the group scheme A_p, the kernel of multiplication by p, is of multiplicative type; equivalently, after extension of scalars it is isomorphic to a power of μ_p.

If $k = \mathbf{F}_q$, if F is the Frobenius endomorphism of A, and if $Pc_A(F; x)$ is its characteristic polynomial, these conditions are also equivalent to

(IV) At least half of the roots of $Pc_A(F; x)$ in $\overline{\mathbf{Q}}_p$ are p-adic units. Equivalently, if $n = \dim A$, the reduction modulo p of $Pc_A(F; x)$ is not divisible by x^{n+1}.

3. The canonical lift

Let A be an ordinary abelian variety over $\overline{\mathbf{F}}_p$. Let $\widetilde{A}$ denote the Serre–Tate canonical lift [4] of A over W. Recall that $\widetilde{A}$ depends functorially on A and is characterized by the fact that the p-divisible group $T_p(\widetilde{A})$ over W attached to $\widetilde{A}$ [5] is the product of the p-divisible groups, uniquely determined by 2.(III), lifting respectively the identity component and the largest etale quotient of $T_p(A)$. Equivalently, $\widetilde{A}$ is the unique lift of A to which every endomorphism of A lifts.

Let $T(A)$ be the integral homology of the complex abelian variety A_c obtained from $\widetilde{A}$ and φ by extension of scalars from W to $\mathbf{C}$:

$$T(A) = H_1(\widetilde{A} \otimes_W \mathbf{C}).$$

One knows that $\widetilde{A}$ descends uniquely to $W_0(\overline{\mathbf{F}}_p)$; hence A_c depends only on A and on the restriction of φ to $W_0(\overline{\mathbf{F}}_p)$. The free $\mathbf{Z}$-module $T(A)$ has rank $2 \dim A$, and it is functorial in A. Moreover, if $l \neq p$ is a prime number, there is a functorial identification

$$T(A) \otimes \mathbf{Z}_l = T_l(A). \tag{1}$$

The canonical lift of the dual abelian variety A^* is dual to $\widetilde{A}$; hence $(A_c)^* = \widetilde{A^*}$, and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbf{Z}(1)$:

$$T(A) \otimes T(A^*) \longrightarrow \mathbf{Z}(1). \tag{2}$$

It is essential to use $\mathbf{Z}(1)$ rather than $\mathbf{Z}$ in order to obtain a theory invariant under complex conjugation. The pairings (2) are compatible, via (1), with the pairings

$$T_l(A) \otimes T_l(A^*) \longrightarrow \mathbf{Z}_l(1).$$

A morphism $\phi : A \to A^*$ defines a polarization of A if and only if the lifted morphism

$$\widetilde{\phi} : \widetilde{A} \to \widetilde{A^*}$$

defines a polarization of A_c.

Put

$$T'_p(A) = \mathrm{Hom}(\mathbf{Q}_p/\mathbf{Z}_p, A(\overline{\mathbf{F}}_p))$$

and

$$T''_p(A) = \mathrm{Hom}_{\mathbf{Z}_p}(T'_p(A^*), \mathbf{Z}(1) \otimes \mathbf{Z}_p).$$

These $\mathbf{Z}_p$-modules are covariant functors of A.

By the definition of the canonical lift, the p-divisible group $T_p(\widetilde{A})$ is the sum of the constant pro-etale group $T'_p(A)$ and the Cartier dual of $T'_p(A^*)$.[1] For a morphism $u : A \to B$, the induced morphism $u : T_p(A) \to T_p(B)$ identifies with the sum of

$$u|T'_p(A) : T'_p(A) \to T'_p(B)$$

and the Cartier transpose of

$${}^t u|T'_p(B^*) : T'_p(B^*) \to T'_p(A^*).$$

Over $\mathbf{C}$ there is a canonical isomorphism

$$\mathbf{Z}(1)/(p^n) \simeq \boldsymbol{\mu}_{p^n},$$

hence an isomorphism of functors

$$T_p(A) = T(A) \otimes \mathbf{Z}_p = T'_p(A) \oplus T''_p(A). \tag{3}$$

4. Finite subgroups and lattices

Recall that if $\varphi : X \to Y$ is an isogeny of complex abelian varieties, the homotopy exact sequence reduces to a short exact sequence

$$0 \to H_1(X) \to H_1(Y) \to \mathrm{Ker}(\varphi) \to 0.$$

The quotient abelian varieties of X by a finite subgroup, and the finite subgroups themselves, correspond bijectively to the overlattices of $H_1(X)$ in $H_1(X) \otimes \mathbf{Q}$.

Let A be an ordinary abelian variety over $\overline{\mathbf{F}}_p$. If n is prime to p, the finite subgroup schemes of order n of A, of $\widetilde{A}$, and of A_c correspond bijectively, and they also correspond to the overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p = T'_p(A) \otimes \mathbf{Q}_p, \qquad V''_p(A) = T''_p(A) \otimes \mathbf{Q}_p.$$

The finite subgroup schemes of A of order p^k are products of an etale subgroup and an infinitesimal subgroup. The etale subgroups of order p^k of A correspond to those subgroups of order p^k of A_c such that the corresponding overlattice R of $T(A)$ is contained in

$$T'_p(A) + V''_p(A).$$

By duality, the infinitesimal subgroups of A correspond to overlattices R of $T(A)$ which are p-isogenous to $T(A)$, i.e. for which $[R : T(A)]$ is a power of p, and which are contained in

$$T''_p(A) + V'_p(A).$$

Altogether, the finite subgroups of A, or the quotient abelian varieties of A, correspond bijectively to the overlattices R of $T(A)$ such that

$$R \otimes \mathbf{Z}_p = (R \otimes \mathbf{Z}_p \cap V'_p) + (R \otimes \mathbf{Z}_p \cap V''_p). \tag{4}$$

[1] We write T_p for the p-adic Tate module and T'_p for the etale component of the p-divisible group.

5. Relative Frobenius

In particular, $A^{(p)}$, which is the quotient of A by the largest infinitesimal subgroup of A killed by p when A is ordinary, is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p-isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbf{Z}_p = T'_p(A) + V''_p(A).$$

6. Descent to $\mathbf{F}_q$

Let A be an abelian variety over $\mathbf{F}_q$, and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined by the pair $(\bar{A}, F)$ obtained from (A, F) by extension of scalars from $\mathbf{F}_q$ to $\overline{\mathbf{F}}_q$. The endomorphism F of $\bar{A}$ factors through the relative Frobenius

$$\mathrm{Fr}(q) : \bar{A} \to \bar{A}^{(q)}$$

and an isomorphism

$$F' : \bar{A}^{(q)} \to \bar{A}.$$

If A is ordinary, let $T(A)$ denote the $\mathbf{Z}$-module $T(\bar{A})$ endowed with the endomorphism F induced by the Frobenius endomorphism of $\bar{A}$. By 5, by the preceding discussion, and by (3), the lattices $T(A)$ and $F(T(A))$ are p-isogenous and one has

$$F(T'_p(A)) = T'_p(A), \tag{5}$$

$$F(T''_p(A)) = q\, T''_p(A). \tag{6}$$

7. The equivalence of categories

Theorem 1. *The functor*

$$A \longmapsto (T(A), F)$$

is an equivalence from the category of ordinary abelian varieties over $\mathbf{F}_q$ to the category of finite-rank free $\mathbf{Z}$-modules T equipped with an endomorphism F satisfying the following conditions:

(a) *F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$.*
(b) *At least half of the roots in $\overline{\mathbf{Q}}_p$ of the characteristic polynomial of F are p-adic units. Equivalently, if T has rank d, the reduction modulo p of $Pc_T(F; x)$ is not divisible by $x^{[d/2]+1}$.*
(c) *There exists an endomorphism V of T such that $FV = q$.*

If condition (a) holds, conditions (b) and (c) are equivalent to

(d) *The module $T \otimes \mathbf{Z}_p$ admits a decomposition, stable under F, into two $\mathbf{Z}_p$-submodules T'_p and T''_p of equal dimension, such that $F|T'_p$ is invertible and $F|T''_p$ is divisible by q.*

Proof. (A) We first prove that (a)+(b)+(c) imply (d). If α is a complex eigenvalue of F, then $\bar{\alpha}$ is another eigenvalue, with the same multiplicity, and

$$\alpha\bar{\alpha} = q.$$

Except for those eigenvalues equal to $\pm q^{1/2}$, the eigenvalues of F in $\mathbf{C}$, hence in $\overline{\mathbf{Q}}_p$, arrange themselves in pairs α and q/α. The two roots α and q/α cannot both be p-adic units. It follows from (b) that $\pm q^{1/2}$ is not an eigenvalue of F, that half of the eigenvalues of F in $\overline{\mathbf{Q}}_p$ are p-adic units, say

$$\alpha_1, \dots, \alpha_{d/2},$$

and that the other half are

$$\beta_1 = q/\alpha_1, \dots, \beta_{d/2} = q/\alpha_{d/2}.$$

Let

$$T_p = T \otimes \mathbf{Z}_p, \qquad V_p = T \otimes \mathbf{Q}_p.$$

Let V'_p be the kernel of $\prod_i (F - \alpha_i)$ on V_p, and let V''_p be the kernel of

$$\psi = \prod_i (F - \beta_i).$$

Then

$$V_p = V'_p \oplus V''_p.$$

Let T'_p be the projection of T_p onto V'_p, and put $T''_p = T_p \cap V''_p$. Since ψ kills V'_p and preserves T, it sends T_p into $T_p \cap V''_p \subset T''_p$. On the other hand,

$$\det(\psi|V'_p) = \prod_{i,j}(\alpha_i - \beta_j)$$

is a p-adic unit. Hence $\psi(T_p) = T_p$ and $T_p \cap V'_p = \psi T_p$, so

$$T_p = T'_p \oplus T''_p.$$

(B) Full faithfulness. Let A and B be abelian varieties over $\mathbf{F}_q$, and consider

$$\psi : \mathrm{Hom}(A, B) \to \mathrm{Hom}_{\mathbf{Z}}(T(A), T(B)).$$

By Tate's theorem [7] and (1), the map

$$\psi_l : \mathrm{Hom}(A, B) \otimes \mathbf{Z}_l \to \mathrm{Hom}_{\mathbf{Z}}(T(A), T(B)) \otimes \mathbf{Z}_l$$

is an isomorphism for $(l, p) = 1$. Thus $\psi \otimes \mathbf{Q}$ is an isomorphism. The group $\mathrm{Hom}(A, B)$ is torsion-free, so ψ is injective. Let $u : A \to B$ be a morphism such that $T(u)$ is divisible by n. Then the induced morphism

$$u_c : A_c \to B_c$$

is divisible by n, hence so is

$$\widetilde{u} : \widetilde{A} \to \widetilde{B}$$

over the generic point of W. The kernel of multiplication by n is flat over W; since $\widetilde{u}$ vanishes on this kernel, both $\widetilde{u}$ and u are divisible by n. Therefore ψ is bijective.

(C) Necessity. The fact that $(T(A), F)$ satisfies (a) follows from Weil. Condition (d), which implies (b) and (c), follows from 6.

(D) Isogenies. Let (T_0, F) satisfy (a)–(d), and let T be a lattice in $T_0 \otimes \mathbf{Q}$, stable under F and still satisfying (d). Suppose that (T_0, F) is the image of an abelian variety $\bar{A}$ over $\overline{\mathbf{F}}_q$; we prove that (T, F) comes from an isogenous abelian variety. Replacing T by $\frac{1}{n}T$, which is isomorphic to it, we may assume $T \supset T_0$. Condition (d) implies that T satisfies (4). Thus T defines a subgroup H of $\bar{A}$, defined over $\mathbf{F}_q$, and

$$(T, F) = T(\bar{A}/H).$$

(E) Essential surjectivity. The functor T induces a functor $T_{\mathbf{Q}}$ from the category of ordinary abelian varieties over $\mathbf{F}_q$ up to isogeny to the category of finite-dimensional $\mathbf{Q}$-vector spaces endowed with an automorphism F satisfying (a) and (b). By (D), it is enough to prove that $T_{\mathbf{Q}}$ is essentially surjective. It is even enough to show that every simple object (V, F) of the image category is reached. By Honda [1], see also [6], there exists an abelian variety $\bar{A}$ over $\overline{\mathbf{F}}_q$ such that the characteristic polynomial of the Frobenius F_A of A is a power of the characteristic polynomial of F. The third characterization in 2 of ordinary abelian varieties shows that $\bar{A}$ is ordinary. Moreover $(T(A) \otimes \mathbf{Q}, F)$ is a sum of copies of (V, F). By (B), the abelian variety up to isogeny $A \otimes \mathbf{Q}$ is a sum of copies of an abelian variety B such that

$$T(B) \otimes \mathbf{Q} = (V, F).$$

□

8. The complex structure

Let (T, F) be a pair satisfying the hypotheses of the theorem, and let $2g$ be the rank of T. Let A be the corresponding abelian variety over $\mathbf{F}_q$, and let A_c be the associated complex abelian variety constructed in 3. Then

$$T = H_1(A_c),$$

so $T \otimes \mathbf{R}$ identifies with the Lie algebra of A_c, and as such it is endowed with a complex structure. Following J.-P. Serre, here is how to reconstruct this complex structure in terms of T, F, and the restriction of φ to $W_0(\overline{\mathbf{F}}_p)$.

Proposition 2. *The complex structure just described on $T \otimes \mathbf{R}$ is characterized by the following properties:*

(I) *The endomorphism F is $\mathbf{C}$-linear.*

(II) *Let v be the valuation of the algebraic closure $\overline{\mathbf{Q}}$ of $\mathbf{Q}$ inside $\mathbf{C}$ extending the valuation of $W_0(\overline{\mathbf{F}}_p)$. The valuations of the g eigenvalues of this endomorphism are strictly positive.*

Condition (I) is evident, and condition (II) follows because the action of F on the Lie algebra of $\widetilde{A}$ is congruent to zero modulo p. The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) for (T, F).

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